

Training Configuration and Summary:

Sequencing:

- timesteps: 60
- batch_size: 128

Model Hyperparameters:

- min_units: 128
- max_units: 256
- units_step: 64
- dropout_min: 0.2
- dropout_max: 0.3
- dropout_step: 0.1
- activation_functions: ['tanh']
- optimizers: ['adam', 'Nadam']

Tuning:

- max_epochs: 30
- hyperband_iterations: 3

Directories:

- directory: my_dir
- project_name: bigru_hyperparameter_tuning

BiGRU Training Report

Report Generated: 2025-04-10_06-11-02

Training Start: 2025-04-09 22:02:05

Training End: 2025-04-09 23:32:49

Training Duration: 5443.35 seconds

Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: Nadam

Dropout Rate: 0.20

Units per Layer:

Layer 1: 256 units, Layer 2: 256 units

Performance Metrics:

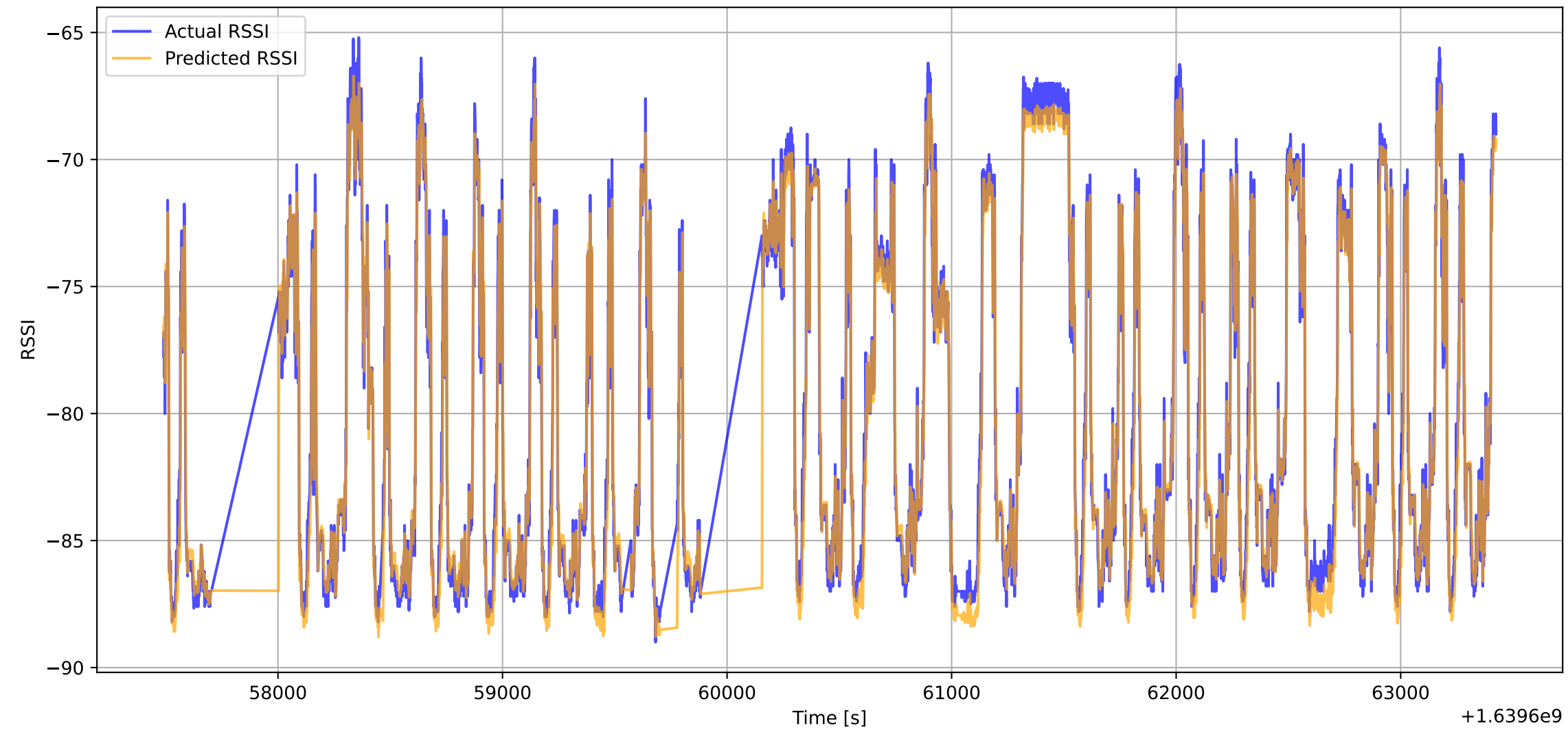
MAE: 0.9660

MSE: 1.8823

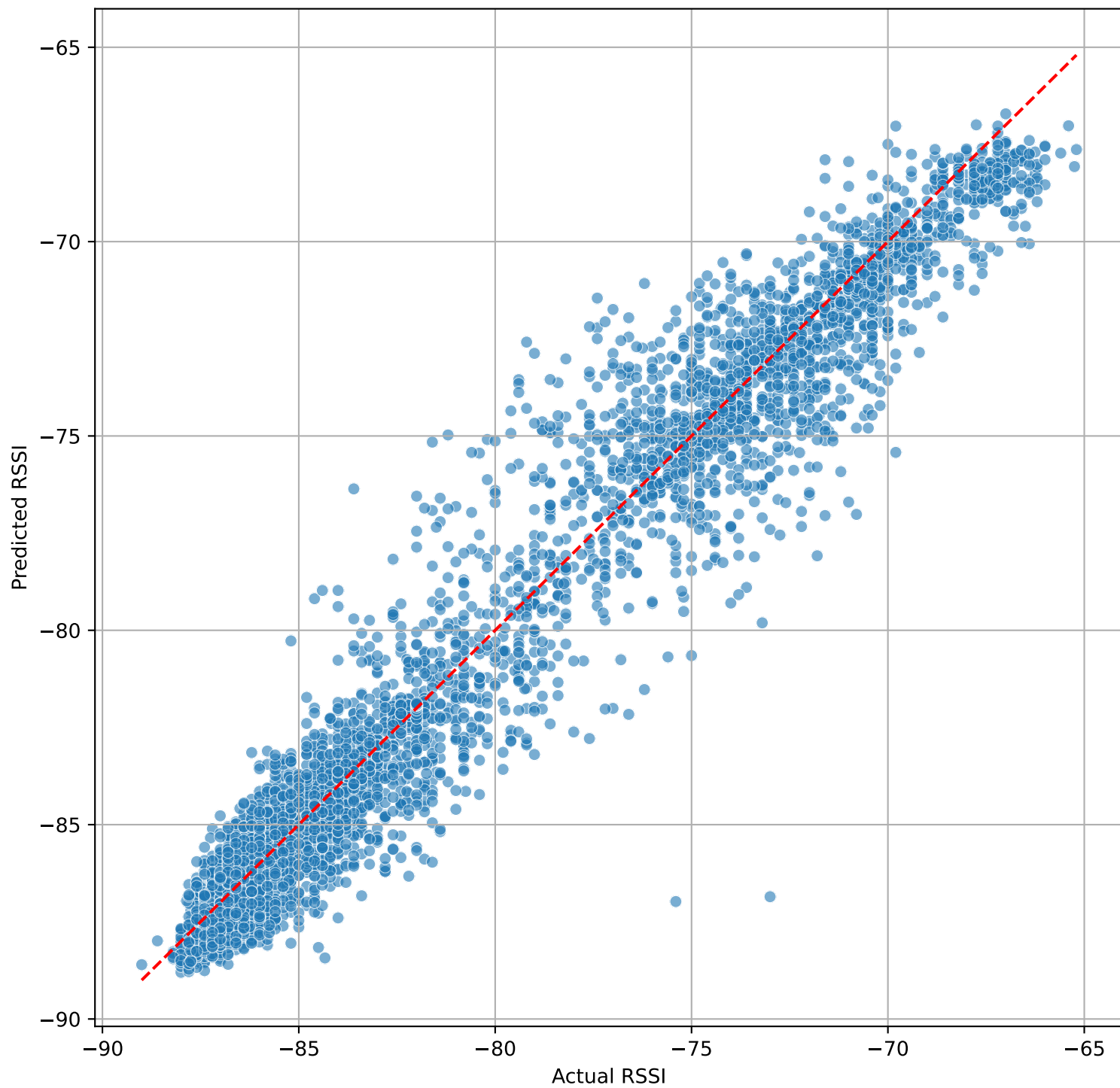
RMSE: 1.3720

R2 Score: 0.9572

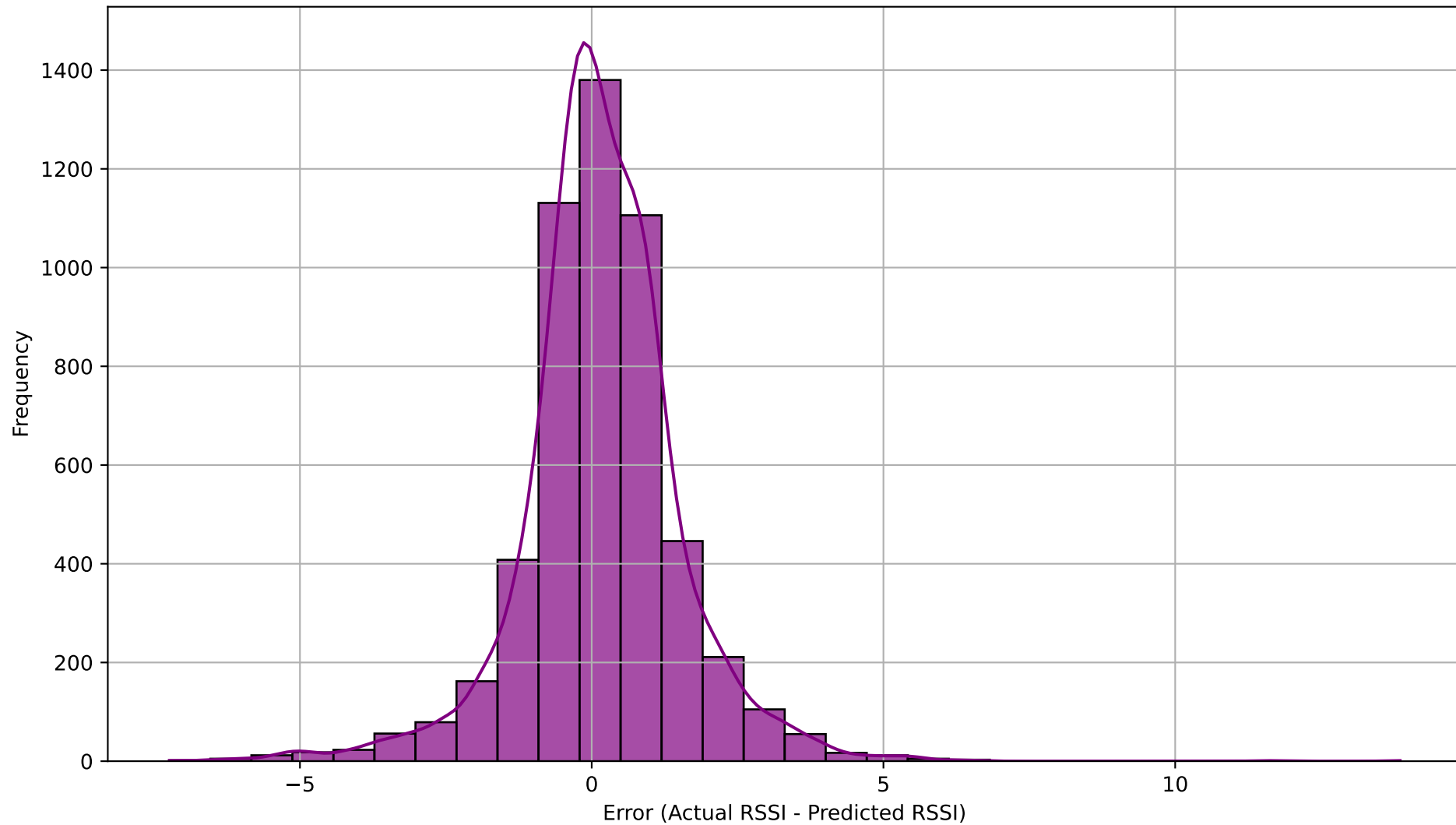
Actual vs Predicted RSSI Over Time



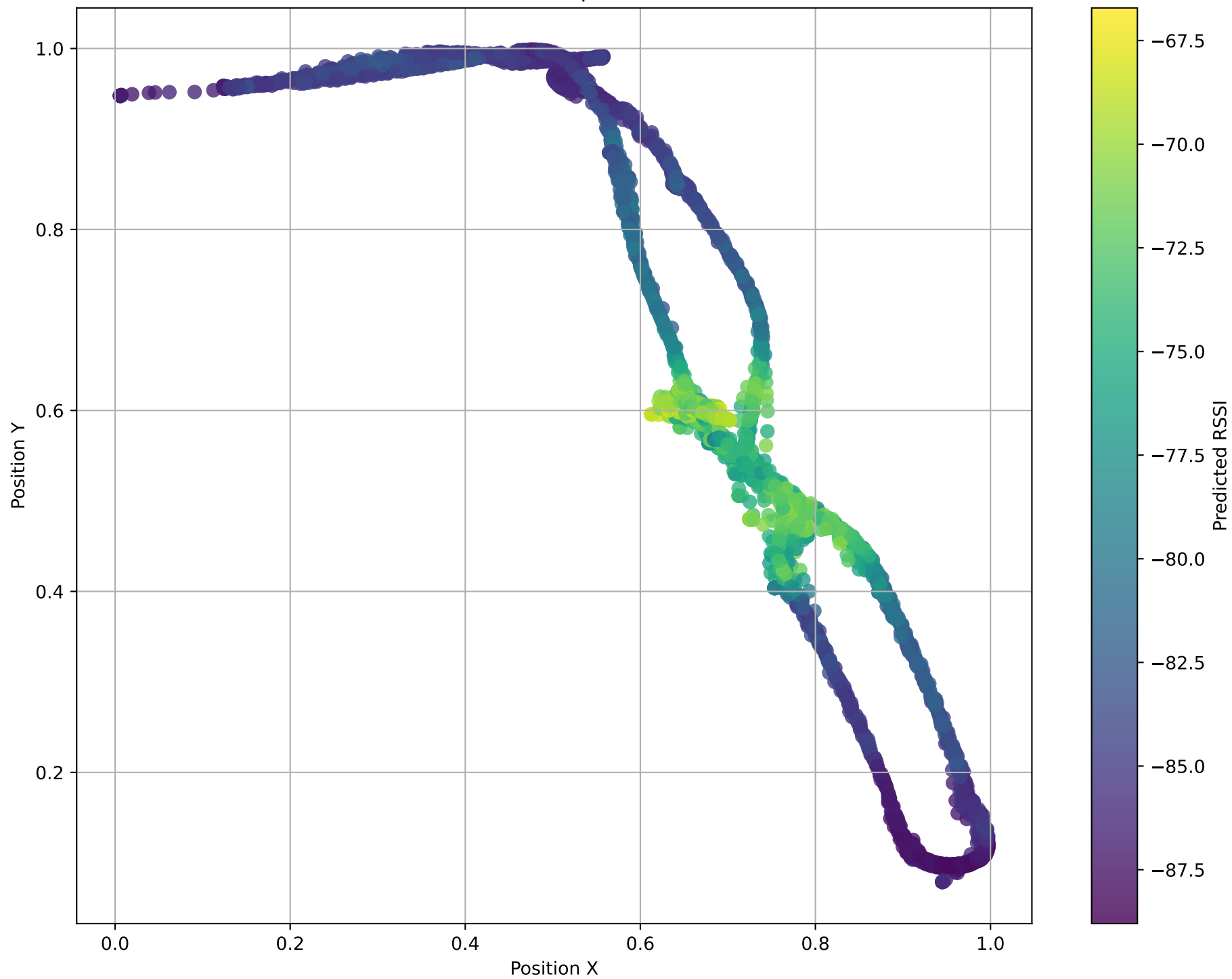
Actual vs Predicted RSSI



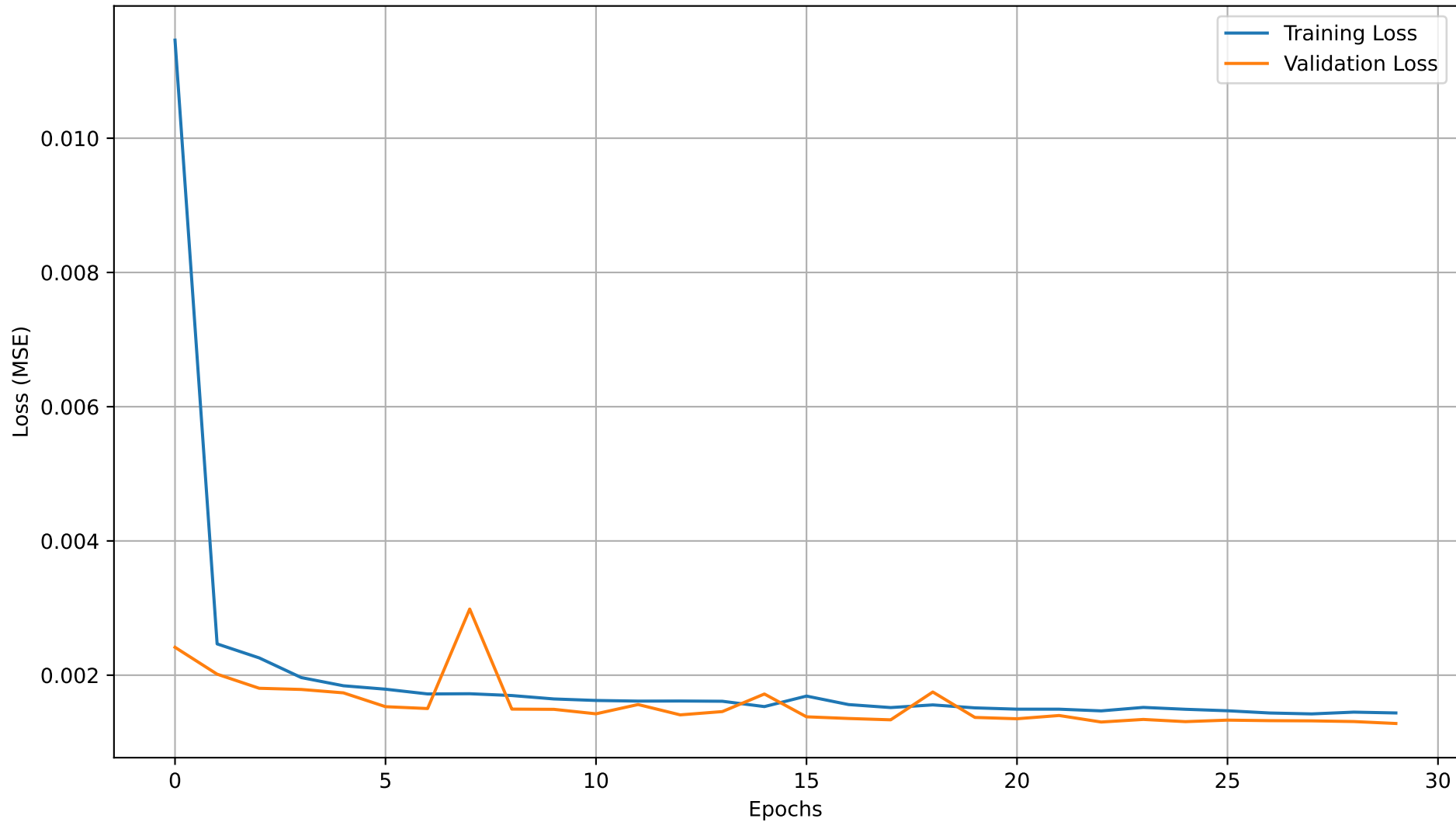
Distribution of Prediction Errors



Predicted RSSI in Spatial Context



Training and Validation Loss Over Epochs



Training and Validation MAE Over Epochs

